

RAGnRecipe

A Medically Aware Recipe Generator

I. Introduction

In the age of personalized health and AI, bridging the gap between medical knowledge and everyday lifestyle choices is critical. Large Language Models (LLMs) offer a promising solution, especially when tailored for specific domains. In this project, we present RAGnRecipe, a medically aware recipe generation system. It combines domain-specific retrieval techniques with a dual Retrieval-Augmented Generation (RAG) framework to provide dietary recommendations based on an individual's health profile.

Our goal was to build an intelligent system capable of interpreting medical histories and generating recipes that not only align with nutritional needs but also respect medical constraints (e.g., diabetes, hypertension). To accomplish this, we integrated structured medical facts, health-annotated recipe datasets, and a two-tiered RAG pipeline. The final generation is powered by the Mistral 7B Instruct model, chosen for its strong instruction-following capabilities and open-access usability.

II. Dataset Description

1. **Recipe Dataset** We used a Kaggle-hosted dataset containing over 2 million recipes, annotated with:

- Ingredients
- Cooking Instructions

This massive corpus enabled broad coverage across global cuisines and dietary patterns. A subset of this data was sampled out and additional tags were created to identify which medical conditions these recipes cater to or are good for, using rule-based heuristics.

2. **Medical Facts Database** To explain *why* a recipe is suitable for a given condition, we created a JSONL corpus covering 50 of the most commonly used ingredients. Each ingredient is associated with:

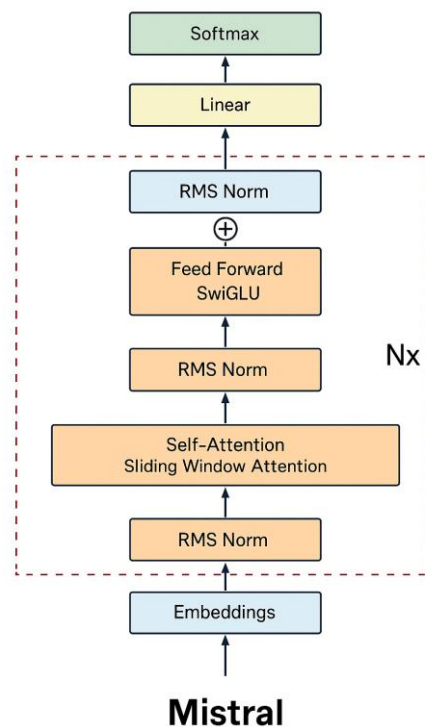
- 3–5 medical facts
- Associated health conditions (e.g., garlic → heart disease, blood pressure)
- Facts generated through a combination of manual curation and GPT-based medical fact synthesis from reputable sources (NIH, WebMD, Healthline)

III. Model Description - Mistral-7B-Instruct-v0.1

Mistral 7B Instruct is a powerful, open-weight language model released by Mistral AI. It builds upon the Mistral 7B base model and has been optimized for instruction-following tasks. With 7 billion parameters, it provides a strong balance between performance and deployability, especially in real-world generative use cases like dialogue systems, summarization, and personalized assistants.

Key architectural features include:

- **Sliding Window Attention:** Mistral 7B uses grouped-query attention with sliding windows to achieve efficient context handling, enabling long-context understanding without incurring large memory overhead.
- **LayerNorm and SwiGLU activations:** The model uses Root Mean Square Layer Normalization (RMSNorm) and SwiGLU activation for better training stability and expressiveness.
- **Instruction-Tuned:** The "Instruct" variant of Mistral 7B is fine-tuned on various datasets tailored for following natural language instructions. It can perform well in zero-shot or few-shot settings without additional task-specific tuning.

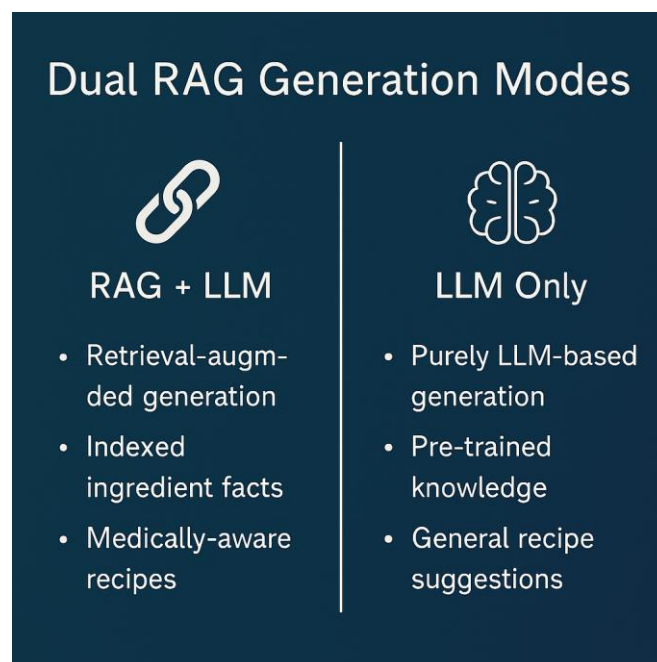


In our system, Mistral 7B Instruct was used to generate final responses that synthesize recipes and their health benefits, based on context retrieved via dual RAG.

IV. Experimental Setup

Steup Overview:

- Pinecone used for vector search and RAG incorporation
- Mistral 7B Instruct was deployed for generation
- Responses were manually evaluated for relevance, clarity, and correctness
- Streamlit was used as the frontend for user interaction



1. Rule-Based Heuristics and Ingredient Tagging:

- Applied rule-based heuristics on the ingredients of each recipe to determine if they are safe for people with conditions such as diabetes, hypertension, gout, anaemia, and more, as well as if they are suitable for goals like weight loss and muscle gain.
- These heuristics were determined by reading resources and articles available online from trusted health websites such as Medline, Food.com and more

```
def Label_Data_Rule_Based(row): 1 usage & Anirudh Rao
try:
    ingredients = eval(row['NER']).lower()
except Exception:
    ingredients = []

ingredient_text = ' '.join(ingredients)

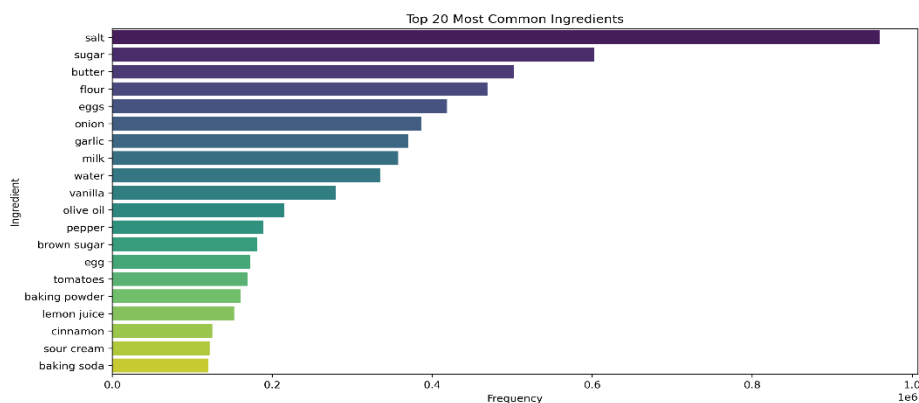
labels = {
    'diabetes_safe': True,
    'hypertension_safe': True,
    'heart_disease_safe': True,
    'pregnancy_safe': True,
    'weight_loss': True,
    'muscle_gain': True,
    'gluten_free_safe': True,
    'lactose_free_safe': True,
    'kidney_disease_safe': True,
    'ibs_safe': True,
    'cholesterol_friendly': True,
    'pcos_friendly': True,
    'gout_safe': True,
    'anemia_support': False,
    'thyroid_friendly': True,
    'acid_reflux_safe': True
}

# Diabetes
if any(k in ingredient_text for k in ['sugar', 'honey', 'corn syrup', 'sweetened', 'molasses', 'rice', 'noodles']):
    labels['diabetes_safe'] = False

# Hypertension & Heart disease
if any(k in ingredient_text for k in ['salt', 'soy sauce', 'bacon', 'processed meat', 'sausage']):
    labels['hypertension_safe'] = False
    labels['heart_disease_safe'] = False
```

2. Ingredient Frequency Analysis and Fact Curation:

- Analyzed the frequency of ingredients in the dataset to identify the 50 most common ones.
- Generated or curated 3–5 medical facts for each of these ingredients and linked them to relevant health conditions.

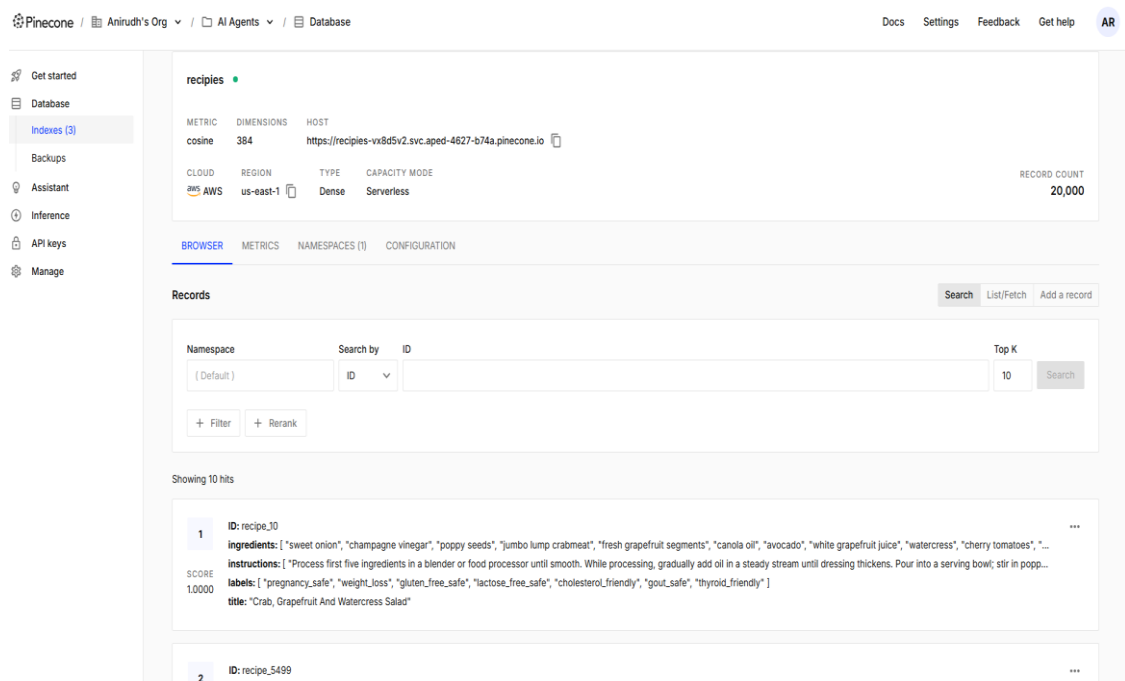
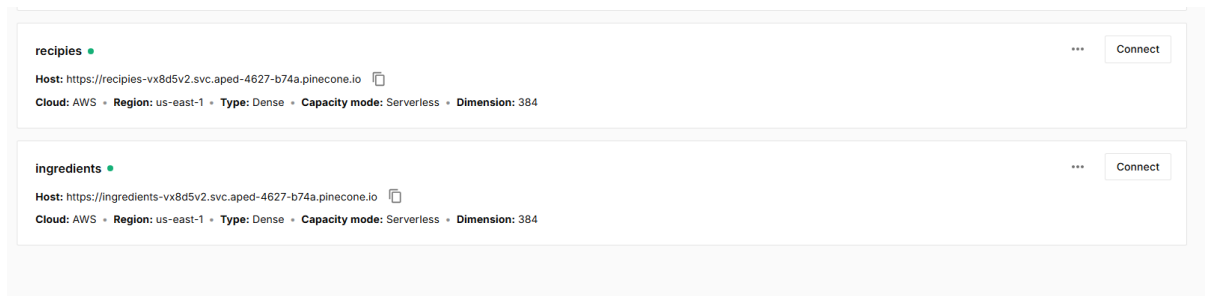


3. Dual Retrieval-Augmented Generation (RAG) Pipeline

- Vector DB: Pinecone** – We utilized Pinecone as our vector database to store and retrieve embeddings. Pinecone offers a scalable and efficient platform for managing high dimensional vectors enabling rapid similarity searches which is essential for system performance.
- Embedding Model:** After evaluating multiple models we selected Sentence Transformers all-MiniLM-L6-v2 (384 Dimensions) to generate embeddings. This is a **6-layer MiniLM model** fine-tuned using a **contrastive learning objective** on a large dataset of sentence pairs. It's designed to convert sentences or short texts into dense vector embeddings that capture their semantic meaning. It uses **distilled BERT architecture**, making it significantly smaller and faster than larger models.

The reasons for selecting this model over others:

- Fast and Lightweight – Ideal for real time applications
- Strong Performance – Despite the small size it performs competitively on sentence similarity and semantic search tasks
- Low Memory Footprint – Can be used efficiently on CPU and small GPUs and does not take up much space, leaving more memory for the Generation Model

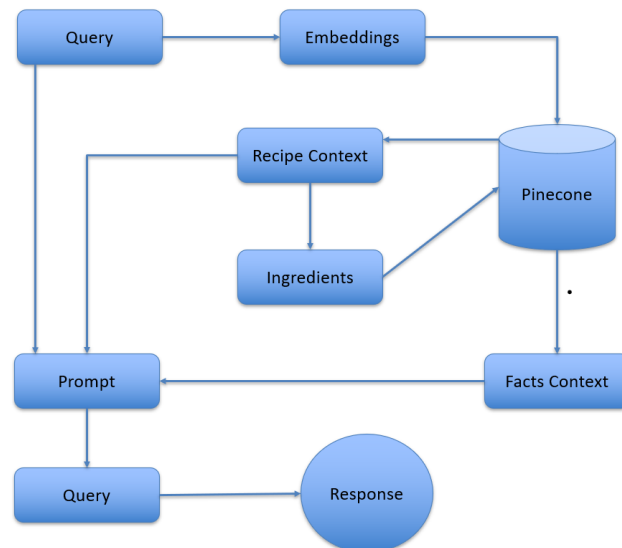


RAG Stage 1: Recipe Retrieval

- User query is embedded and used to retrieve the most semantically relevant recipes.

RAG Stage 2: Medical Fact Retrieval

- Extracted ingredients from RAG Stage 1 recipes
- Each ingredient is queried to retrieve medical benefits and associated conditions from the medical facts ingredients database



V. Results

1. Question : Generate a Recipe with Beef Broccoli and Garlic:

- Rag Response:

```

RAG Response:

Recipe Name: Beef And Broccoli Recipe
Ingredients:
- beef
- garlic
- broccoli
- soy sauce
Instructions:
1. Blend steak sauce, soy sauce and garlic in nonmetal dish.
2. Add in steak, stirring to coat.
3. Cover; chill 30 min, stirring occasionally.
4. Remove steak from marinade; reserve marinade.
5. Cook and stir steak in large, lightly greased skillet over medium-high heat for 3 to 4 min or possibly till meat is no longer pink.
6. Remove steak with slotted spoon; keep hot.
7. Heat vegetables and reserved marinade in same skillet to a boil; reduce heat.
8. Cover; simmer for 2 to 3 min.
9. Stir in beef.
10. Serve over warm cooked rice.
11. Special Extra: Add in 1/2 to 1 teaspoon warm pepper sauce with soy sauce.

Ingredient-Level Medical Facts:
- beef: It provides significant amounts of iron, which helps prevent iron-deficiency anemia by aiding in oxygen transport in the blood.
- beef: It contains B vitamins, including B12 and niacin, which support energy metabolism and nervous system health.
- beef: Beef is an excellent source of high-quality protein, containing all essential amino acids necessary for muscle growth and repair.
- garlic: Garlic has been shown to boost the immune system, helping the body fight off illnesses like the common cold.
- garlic: Garlic consumption may help lower blood pressure and cholesterol levels, reducing the risk of cardiovascular diseases.
- garlic: Garlic is rich in antioxidants that protect against cell damage and aging, and may reduce the risk of Alzheimer's disease and dementia.
- broccoli: Broccoli contains fiber, aiding in digestion and promoting gut health.
- broccoli: It provides vitamin K and calcium, essential for bone health.
- broccoli: High in vitamin C, broccoli supports immune function and skin health.

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- LLM Only Response:

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LLM-Only Response:
5. Nutritional Information (calories, protein, fat, carbs)
6. Servings
7. Time to Prepare/Cook
8. Health Benefits of Ingredients
9. Optional Additions or Substitutions

Recipe 1: Beef Stir Fry with Broccoli and Garlic

Ingredients:
1. 1 lb lean beef sirloin, cut into thin strips
2. 2 cups broccoli florets
3. 3 cloves garlic, minced
4. 1 tbsp olive oil
5. Salt and pepper to taste
6. Soy sauce for serving

Instructions:
1. Heat olive oil in a large skillet over medium-high heat.
2. Season beef strips with salt and pepper.
3. Add beef to the skillet and cook until browned, about 5 minutes per side. Remove from skillet and set aside.
4. In the same skillet, add broccoli florets and minced garlic. Cook until tender, about 5 minutes.
5. Return beef to the skillet and stir well to combine.
6. Serve hot with soy sauce for dipping.

Medical Explanation:
Beef is an excellent source of high-quality protein which helps build and repair muscles and tissues. It also contains essential vitamins such as B vitamins, zinc, and iron. Broccoli is rich in antioxidants, fiber, and vitamin C which supports immune function and healthy digestion. Garlic has anti-inflammatory properties and may help lower blood pressure and cholesterol levels. This dish is low in calories and can be a great option for weight management and heart health.

Nutritional Information:
Calories: 280 | Protein: 25g | Fat: 10g | Carbs: 10g
Servings: 4 | Time to Prepare/Cook: 20 minutes
Health Benefits of Ingredients:
Beef - Supports muscle growth and repair, provides essential nutrients like B vitamins, zinc, and iron.
Broccoli - Rich in antioxidants, fiber, and vitamin C which supports immune function and healthy digestion.
Garlic - Has anti-inflammatory properties and may help lower blood pressure and cholesterol levels.
Optional Additions or Substitutions:
For added flavor and nutrition, you could try adding sliced bell peppers or mushrooms to the stir fry. You could also substitute the beef with tofu or tempeh for a vegetarian option.
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We can see that the Rag Response has generated a recipe with detailed Instructions and more specific Ingredient-level Medical Facts as compared to the LLM only Response. The LLM only response however does provide some additional information on Macronutrients obtained for this recipe which highlights some additional improvement that is possible for our current set up

2. Question : Weight loss recipe for a person with Diabetes



RAGnRecipe

Get recipes tailored to medical needs with ingredient-level facts.

Enter your query:

Weight loss recipe for a person with diabetes

Generate

RAG + LLM Response

Recipe Name: Diabetic Fruit Bars Recipe Ingredients:

- vanilla
- eggs
- baking soda
- water
- raisins
- dates
- prunes
- flour
- nuts
- butter
- salt

Instructions:

- Coarsely chop dates, raisins and prunes.

LLM-Only Response

- Calories per Serving
- Time to Prepare
- Nutritional Information (protein, carbs, fats, etc.)
- Health Benefits of the Recipe
- Any Additional Tips or Recommendations

Recipe 1: Grilled Chicken and Veggie Bowl Ingredients:

- Boneless chicken breast (4 oz)
- Bell pepper (1 medium)
- Zucchini (1 medium)
- Red onion (1/2 cup)
- Olive oil (1 tbsp)
- Salt (to taste)
- Black pepper (to taste)
- Garlic powder (1 tsp)

Instructions: - Coarsely chop dates, raisins and prunes. - Put in a small pan, add in 1 c. water and bring to a boil. - Cook for 1 minute. - Put fruits and juices in a mixing bowl. - Let cold. - Then add in butter, Large eggs, baking soda, vanilla and salt and mix well. - Stir in flour a little at a time. - Mix in nuts and spread batter in a greased 9 inch square pan. - Bake in a 350 degree oven 25 to 30 min. - Cold and cut in squares. Ingredient-Level Medical Facts: - butter: Butter contains vitamin A, which is essential for immune function, vision, and skin health. - butter: Grass-fed butter contains higher levels of omega-3s and fat-soluble vitamins. - butter: Butter contains butyrate, a short-chain fatty acid that supports colon health and reduces inflammation. Recipe Name: Sugarless Applesauce Cake(Diabetic Recipe) Ingredients: - unsweetened applesauce - vanilla extract - baking soda - ground nutmeg - eggs - apples	- Salt (to taste) - Black pepper (to taste) - Garlic powder (1 tsp) - Paprika (1 tsp) Cooking Instructions: - Preheat grill to medium heat. - Cut bell pepper, zucchini, and red onion into bite-sized pieces. - In a bowl, mix olive oil, salt, black pepper, garlic powder, and paprika together. Add chicken and vegetables to the mixture and toss until evenly coated. - Place chicken and vegetables onto the grill and cook for about 5 minutes on each side, or until chicken is cooked through and vegetables are tender. - Serve hot and enjoy! Medical Explanation: This recipe is low in carbohydrates and high in protein, making it an excellent choice for someone with diabetes. The use of spices like garlic powder and paprika can also help regulate blood sugar levels by stimulating insulin production. Additionally, this dish contains healthy fats from olive oil which can improve cholesterol levels and reduce inflammation. Calories per Serving: Approximately 200 calories Time to Prepare: About 20 minutes Nutritional Information: Protein - 25g, Carbs - 10g, Fats - 10g Health Benefits of the Recipe: Helps manage weight, improves heart health, reduces risk of type 2 diabetes Additional Tips or Recommendations: You could add more veggies such as broccoli or spinach for extra nutrients. Also, consider using leaner cuts of meat if desired.
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Recipe Name: Meat Pinwheels(Diabetic Recipe) Ingredients: - horseradish - cream cheese - ham - turkey - worcestershire sauce Instructions: - Blend cream cheese, onion, horseradish and Worcestershire sauce until smooth and spreadable. - Separate the slices of ham and turkey, laying one slice on each slice of ham and then spread with 1 tablespoon cheese mixture. - Roll and fasten with toothpicks. Chill. - Just before serving, slice in 1/2-inch slices. - Makes approximately 50 pinwheels. - One roll equals 1 meat. Ingredient-Level Medical Facts: - turkey: Low in saturated fat, making it a heart-healthy alternative to red meats. - turkey: Turkey is a lean source of high-quality protein, essential for muscle growth and repair. - turkey: Contains selenium and zinc, minerals that bolster immune function and thyroid health.	
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We can see that in this case the RAG System outputted 3 recipes with a mixed focus on Diabetes and weight loss. The LLM only response started off with a bit of Additional structured text which was not a part of the prompt but generated a recipe with some sound medical knowledge. This indicates that Recipe and Nutrition Data might have been a part of the Models training corpus which has yielded good baseline results.

3. Question: I need a Muscle building recipe for someone with Anaemia

Get recipes tailored to medical needs with ingredient-level facts.

Enter your query:

I need a Muscle building recipe for someone with Anaemia

Generate

RAG + LLM Response

Recipe Name: Essentials: Spaghetti And Meatballs Recipe Ingredients:

- 1. egg
- 2. tomatoes
- 3. ground nutmeg
- 4. freshly ground black pepper
- 5. directions
- 6. parmesan cheese
- 7. kosher salt
- 8. ground beef
- 9. bread crumbs
- 10. ground pork
- 11. yellow onion
- 12. good red wine
- 13. olive oil
- 14. ground veal
- 15. fresh white breadcrumbs
- 16. parsley
- 17. garlic
- 18. flat-leaf

LLM-Only Response

- 5. Nutritional Information (calories, protein, carbs, fats)
- 6. Time to Prepare and Cook
- 7. Servings
- 8. Optional Additions or Modifications
- 9. Health Benefits of the Recipe

Recipe 1: Chicken Parmesan

- 1. Skinless chicken breast
- 2. Bread crumbs
- 3. Egg whites
- 4. Olive oil
- 5. Tomato sauce
- 6. Mozzarella cheese
- 7. Salt and pepper

Instructions:

- Preheat oven to 375°F.
- In a shallow dish, mix breadcrumbs with salt and pepper.
- In another shallow dish, beat egg whites until stiff peaks form.

- 13. olive oil
- 14. ground veal
- 15. fresh white breadcrumbs
- 16. parsley
- 17. garlic
- 18. flat-leaf
- 19. vegetable oil

Instructions:

- Put the ground meats, both bread crumbs, parsley, Parmesan, salt, pepper, nutmeg, egg, and 3/4 cup warm water in a bowl. Combine very lightly with a fork. Using your hands, lightly form the mixture into 2-inch meatballs. You will have 14 to 16 meatballs.
- Pour equal amounts of vegetable and olive oil into a large (12-inch) skillet to a depth of 1/4 inch. Heat the oil. Very carefully, in batches, place the meatballs in the oil and brown them well on all sides over medium-low heat, turning carefully with a spatula or a fork. This should take about 10 minutes for each batch. Don't crowd the meatballs. Remove the meatballs to a plate covered with paper towels. Discard the oil but don't clean the pan.
- For the sauce, heat the olive oil in the same pan. Add the onion and saute over medium heat until translucent, 5 to 10 minutes. Add the garlic and cook for 1 more minute. Add the wine and cook on high heat, scraping up all the brown bits in the pan, until almost all the liquid evaporates, about 3 minutes. Stir in the tomatoes, parsley, salt, and pepper.
- Return the meatballs to the sauce, cover, and simmer on the lowest heat for 25 to 30 minutes, until the meatballs are cooked through. Serve hot on cooked spaghetti and pass the grated Parmesan cheese.

Ingredient-Level Medical Facts:

- egg: Eggs contribute to satiety and weight management due to their high protein content.
- egg: They are a good source of vitamin B12 and folate, important for red blood cell formation and DNA synthesis.
- egg: They contain choline, a nutrient vital for brain development and liver function.
- egg: Eggs are a complete protein source, providing all nine essential amino acids needed for muscle maintenance and repair.

- 7. Salt and pepper

Instructions:

- Preheat oven to 375°F.
- In a shallow dish, mix breadcrumbs with salt and pepper.
- In another shallow dish, beat egg whites until stiff peaks form.
- Dip each chicken breast in the beaten egg whites, then coat it with the breadcrumb mixture.
- Heat olive oil in an ovenproof skillet over medium heat.
- Add seasoned chicken breasts and cook for 5 minutes per side, until golden brown.
- Remove from skillet and pour tomato sauce over the top of each chicken breast.
- Sprinkle mozzarella cheese over the tomato sauce.
- Place skillet under broiler and bake for 10 minutes, until cheese is melted and bubbly.

Medical Explanation: This recipe is rich in lean protein from the chicken breast, which can help combat anaemia by promoting iron absorption. The breadcrumb coating adds fiber, which also helps with digestion and nutrient absorption. The tomato sauce provides vitamin C, which enhances iron absorption. Finally, the mozzarella cheese contains calcium, which supports bone health and muscle function.

Nutritional Information: Calories: 350 Protein: 30g Carbs: 20g Fats: 15g Time to Prepare and Cook: 30 minutes Servings: 4 Optional Additions or Modifications: You can add vegetables like bell peppers or zucchini to increase the nutrient content of this dish. You can also use gluten-free bread crumbs if you have celiac disease or non-celiac gluten sensitivity. Health Benefits of the Recipe: This recipe promotes healthy muscle growth while providing essential nutrients for combatting anaemia. It is low in saturated fat and high in fiber, making it a great option for those looking to maintain a healthy weight.

****NOTE**** : The model has been allowed to hallucinate with Temperature 0.3 and top_p 0.8 with sampling enabled so the same query does output different results but can be quite similar in many aspects.

VI. Summary and Conclusions

In this project, we introduced **RAGnRecipe**, a medically aware recipe generator that bridges AI and healthcare through a dual Retrieval-Augmented Generation (RAG) system. The tool intelligently interprets user queries in the context of their health conditions and generates recipes aligned with both medical and nutritional guidelines.

We combined structured ingredient-level medical facts with an extensive recipe dataset and used Pinecone for efficient vector-based retrieval. The generation component was powered by **Mistral 7B Instruct**, chosen for its instruction-following capabilities and low resource requirements. Our dual-stage RAG pipeline first retrieves relevant recipes and then augments them with medically grounded ingredient insights.

Our results showed that the RAG-based system consistently generated contextually appropriate recipes with ingredient-specific medical benefits, outperforming standalone LLM responses in clarity and relevance. However, standalone LLMs demonstrated strengths in providing structured nutritional data, indicating future improvements could involve merging the strengths of both paradigms.

In conclusion, **RAGnRecipe** demonstrates a scalable and interpretable approach to personalized recipe generation for users with health considerations, and it opens the door for further research in combining LLMs with structured domain-specific knowledge for health and lifestyle applications.

Future Scope and Improvements

- **Macronutrient Integration:** A future version of the system can integrate nutritional breakdowns (calories, protein, fats, carbs) alongside medical reasoning for a more comprehensive dietary recommendation.
- **Expanded Ingredient Coverage:** Currently, only the 50 most frequent ingredients are annotated. Extending this database to cover hundreds or thousands of ingredients would increase system robustness and dietary flexibility.
- **Personalized Embeddings:** Instead of a generic user query embedding, health profile-aware embeddings could help improve the relevance of recipe retrieval further.
- **Fine-Tuning on Domain Data:** The Mistral 7B model could be further fine-tuned on medical recipes and nutrition data to reduce hallucinations and improve consistency in output.
- **Multi-modal Input:** Future iterations could accept user-uploaded health reports (PDF, scanned prescriptions) or voice-based queries to enhance accessibility.
- **User Feedback Loop:** Incorporating user feedback on generated recipes could enable continual learning and refinement of the generation process.

VII. References

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5. Mavuduru, A. (2024, November 28). *How to Use LLaMA for Text Generation*. Medium. Retrieved from <https://amolmavuduru.medium.com/how-to-use-llama-for-text-generation-4f810d168199>
6. **Streamlit Docs.** (n.d.). *Build a Basic LLM Chat App*. Retrieved from <https://docs.streamlit.io/develop/tutorials/llms/build-conversational-apps>

****Note**** : Initial Development was done using LLAMA 3.2-3B. Hence references contain documents regarding LLAMA models. Information obtained from here was similarity applicable for my Mistral Model

VIII. Codes Developed

1. Preprocessing
 - EDA.py – Used to Identify the Most occurring ingredients in the Recipe Dataset
 - Total Lines: 38
 - Lines From net: 0
 - Modified Lines: 0
 - Added Lines: 38
 - Preprocess_csv.py - Applies rule-based labeling to a recipe's ingredients to determine dietary suitability and health condition compatibility.
 - Total Lines: 159
 - Lines From net: 0
 - Modified Lines: 0
 - Added Lines: 159
2. RAG
 - Generate_rag_data_format.py - Formats a recipe row from the DataFrame into a structured dictionary suitable for Retrieval-Augmented Generation (RAG) ingestion.
 - Total Lines: 94

- Lines From net: 0
- Modified Lines: 0
- Added Lines: 94
- Generate_rag_ingredients.py – Formats an ingredient row into a structured dictionary suitable for use in a Retrieval-Augmented Generation (RAG) system.
 - Total Lines: 65
 - Lines From net: 0
 - Modified Lines: 0
 - Added Lines: 65
- Llm.py – Constructs a structured prompt for the RAG system using a user query, relevant recipes, and associated ingredient-level medical facts. Then using the Query, Context Retrieved generates LLM output
 - Total Lines: 310
 - Lines From net: 260
 - Modified Lines: 30
 - Added Lines: 50
- Sample_upsert.py – Pinecone Ingestion Script for Medically Aware Recipe Generator (RAGnRecipe)
 - Total Lines: 90
 - Lines From net: 80
 - Modified Lines: 10
 - Added Lines: 10
- 3. Streamlit App – app.py – Frontend for my Project
 - Total Lines: 72
 - Lines From net: 40
 - Modified Lines: 20
 - Added Lines: 32

Overall Code %

- Total Lines: 828
- Lines From net: 380
- Modified Lines: 60
- Added Lines: 448
- Percentage: 39%